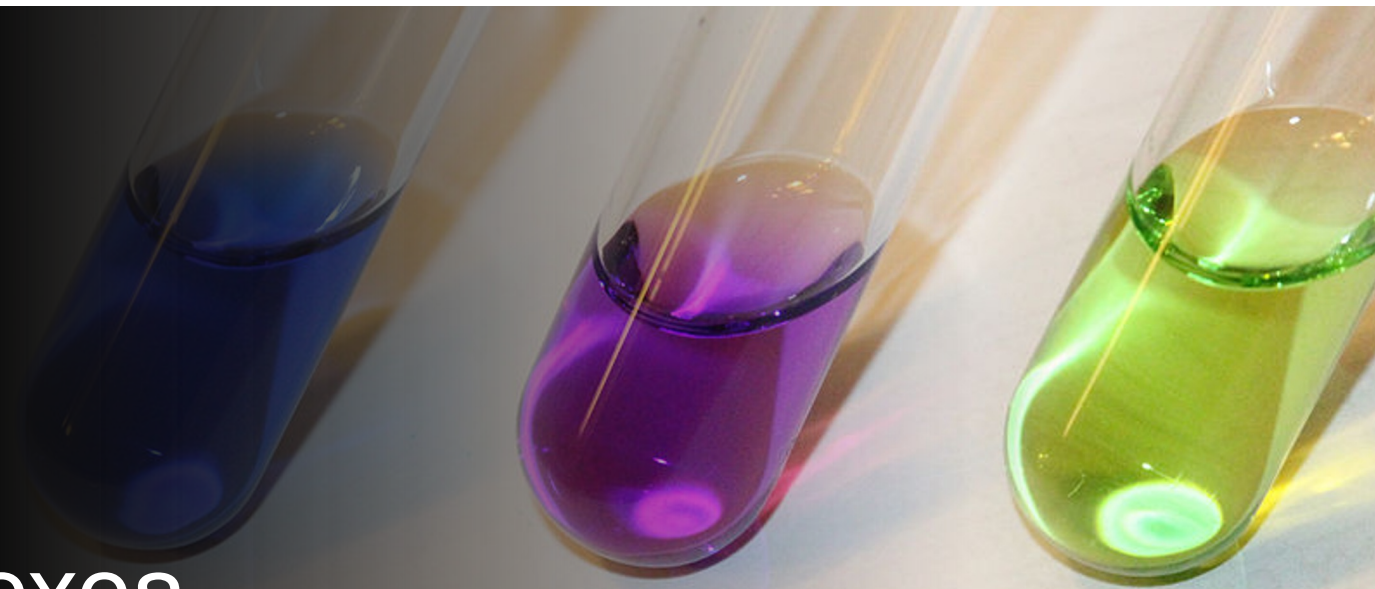


Color in Complexes



Red

HUMANS AND THE MAJORITY OF
OTHER VERTEBRATES



Blue

SPIDERS, CRUSTACEANS, SOME
MOLLUSCS, OCTOPUSES & SQUID



Green

SOME SEGMENTED WORMS, SOME
LEECHES, & SOME MARINE WORMS

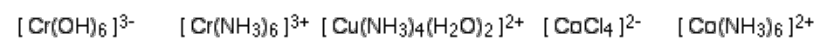
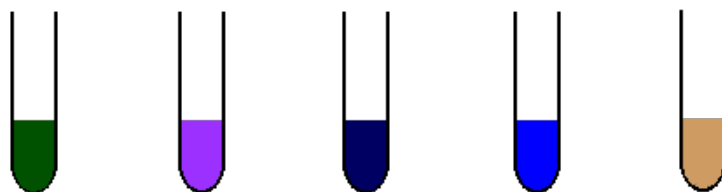
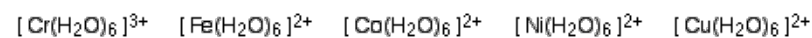
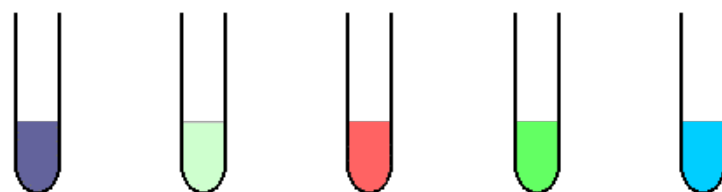


Violet

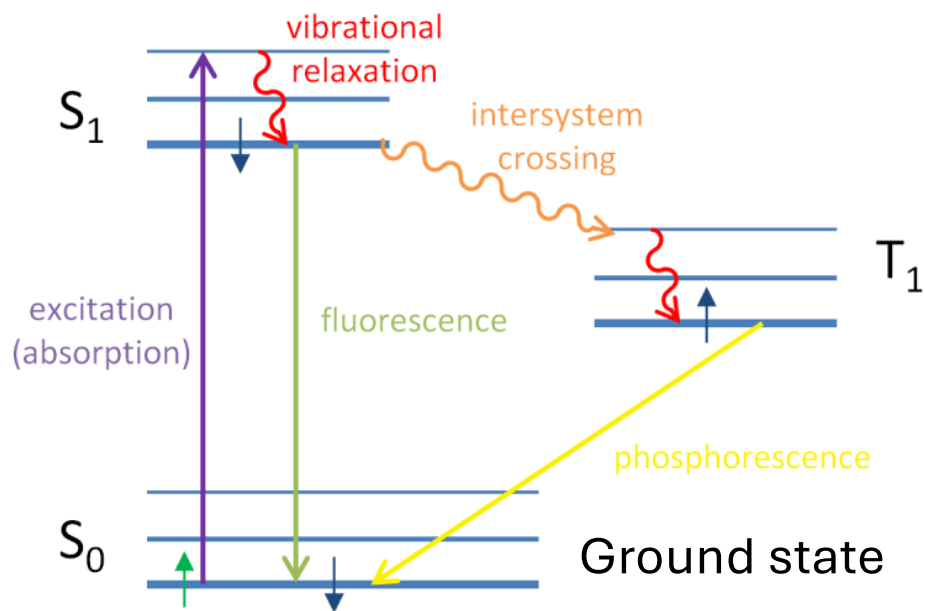
MARINE WORMS INCLUDING PEANUT
WORMS, PENIS WORMS & BRACHIOPODS

3D Transitional metal ions

3	4	5	6	7	8	9	10	11	12
21	22	23	24	25	26	27	28	29	30
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Scandium 44.96	Titanium 47.88	Vanadium 50.94	Chromium 52.00	Manganese 54.94	Iron 55.85	Cobalt 58.93	Nickel 58.69	Copper 63.55	Zinc 65.39



Singlet and triplet excited states



*Jablonski Diagram
(simplified version)*

Energy gap (E_g) between ground state and excited state

$$E_g = \frac{hc}{\lambda}$$

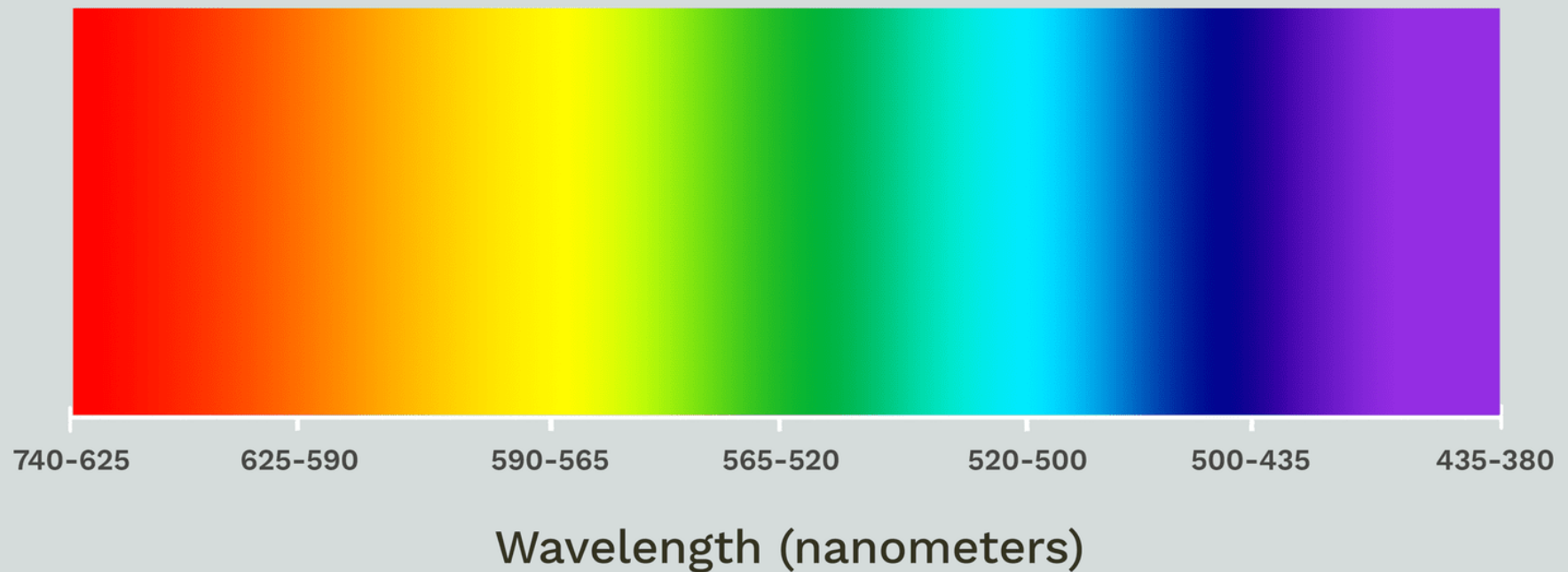
h – Planck's constant
 $\sim 6.626 \times 10^{-34} \text{ m}^2 \text{ kg} / \text{s} (\text{J/Hz})$
 $\sim 4.135 \times 10^{-15} \text{ eV/Hz}$

c – speed of light
 $\sim 3 \times 10^8 \text{ m/sec}$

λ = wavelength

The Visible Light Spectrum

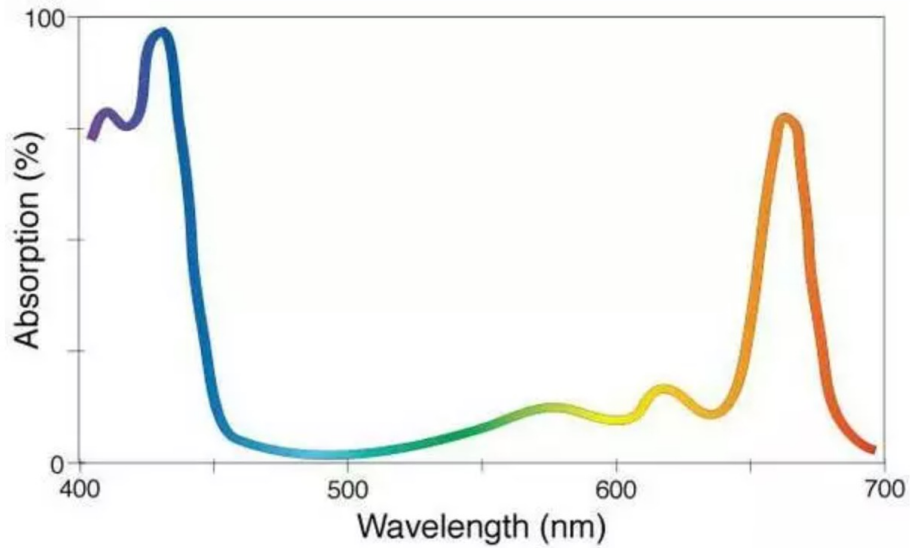
The visible light spectrum is the section of the electromagnetic radiation spectrum that is visible to the human eye.



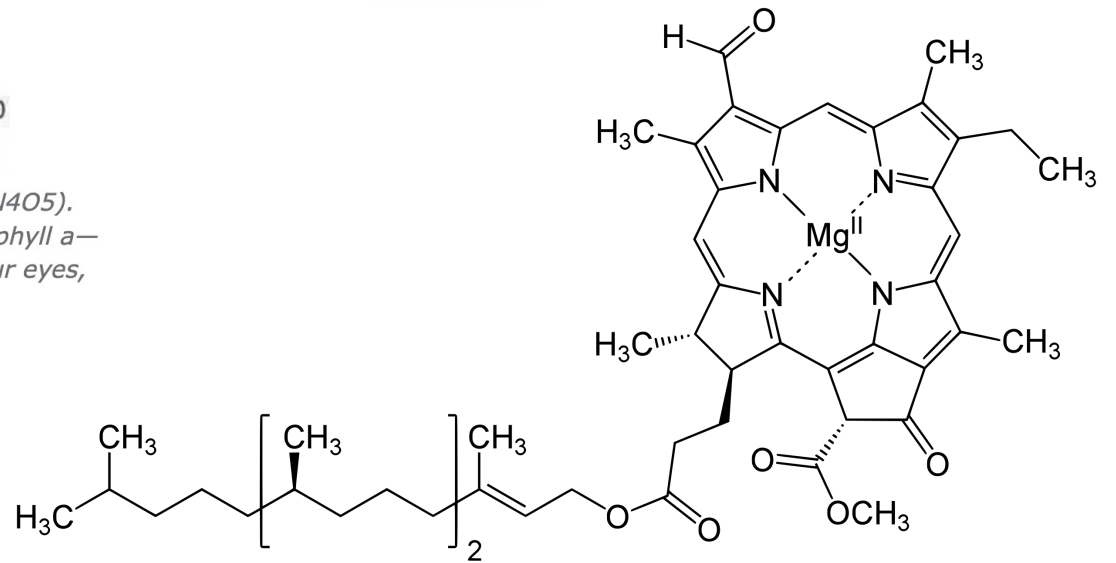
VIBGYOR



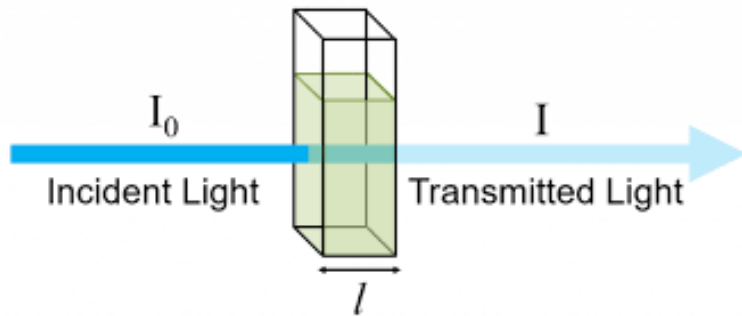
Chlorophyll



The absorption spectrum of the chemical compound chlorophyll a (C₅₅H₇₂MgN₄O₅). Because it absorbs primarily the violet/blue and orange/red wavelengths, chlorophyll a—the substance that is essential to photosynthesis in plants—appears green to our eyes, giving plants their green hue. (Image Source: NASA)



Beer-Lambert Law



Absorbance is directly proportional to the concentration of the absorbing substance and the path length of the light through it

$$A = \epsilon c l,$$

where

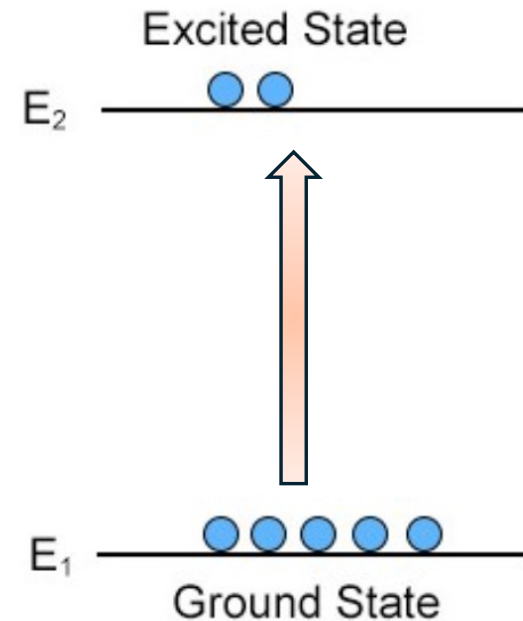
- ϵ is the *molar absorption coefficient* of that material;
- c is the *molar concentration* of those species;
- l is the path length.

Molar absorption coefficient

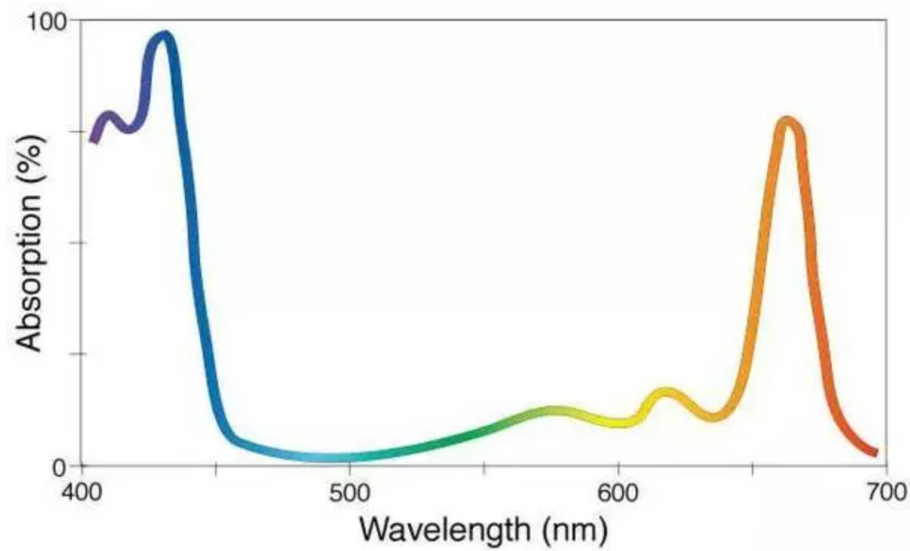
Usually expressed in terms of $\text{M}^{-1}\cdot\text{cm}^{-1}$ or $\text{L}\cdot\text{mol}^{-1}\cdot\text{cm}^{-1}$

Intrinsic property of compound

Basically, how strongly a compound absorbs light



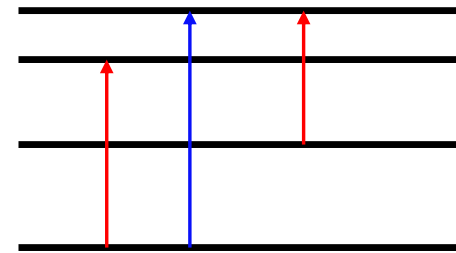
Extinction coefficient directly proportional to the **probability of an electronic transition** occurring between energy levels when a molecule absorbs light



The absorption spectrum of the chemical compound chlorophyll a ($C_{55}H_{72}MgN_4O_5$). Because it absorbs primarily the violet/blue and orange/red wavelengths, chlorophyll a—the substance that is essential to photosynthesis in plants—appears green to our eyes, giving plants their green hue. (Image Source: NASA)

Multiple transitions

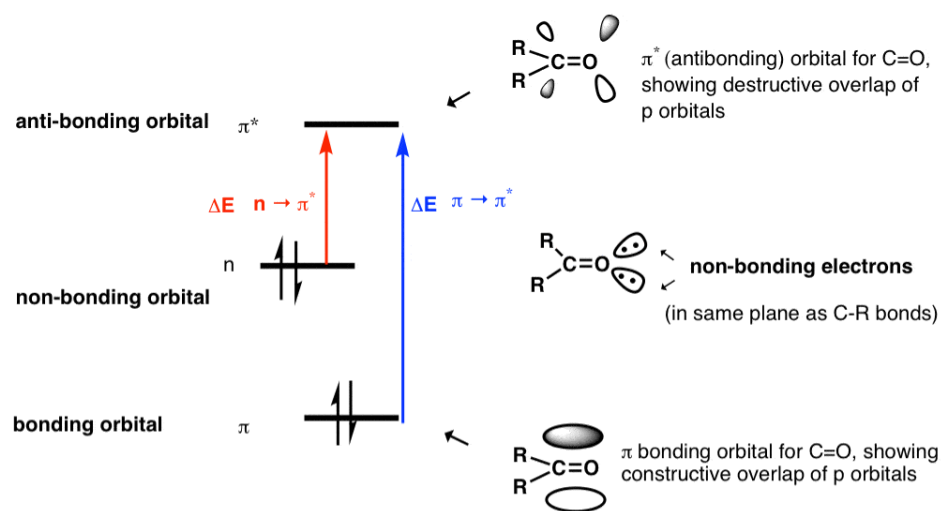
Can you comment on the extinction coefficient?



What will be the probability of such transitions?

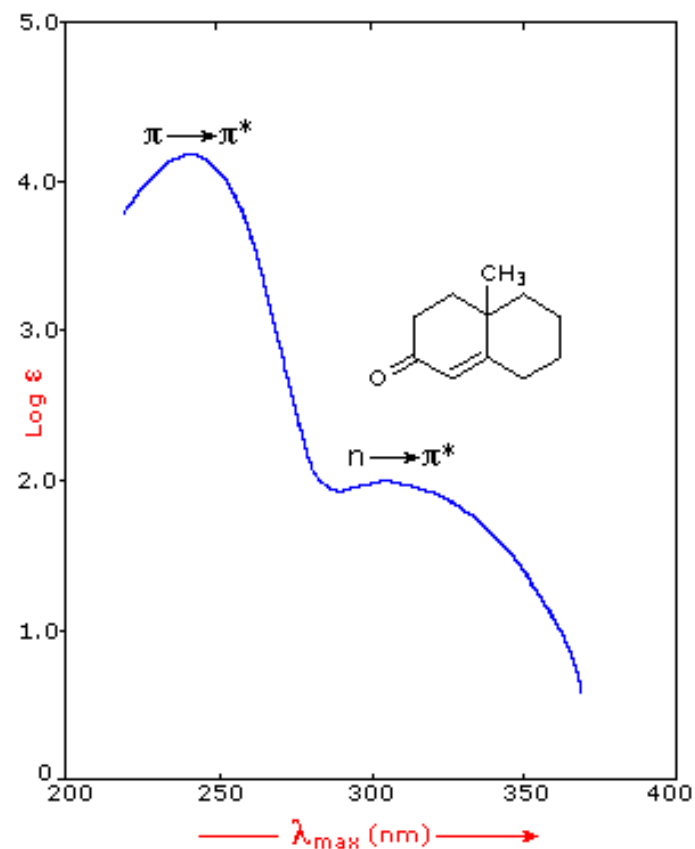
Allowed vs. Forbidden Transitions

Simplified Molecular Orbital Diagram for Acetone (Propanone)



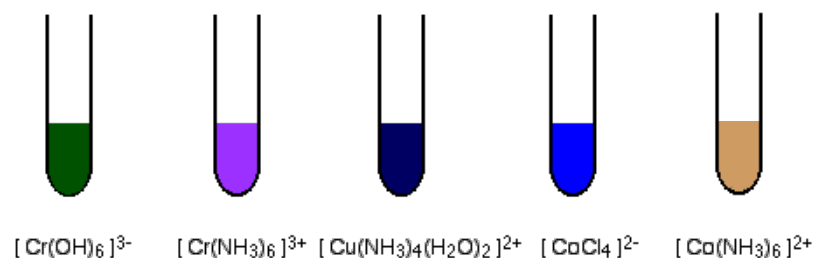
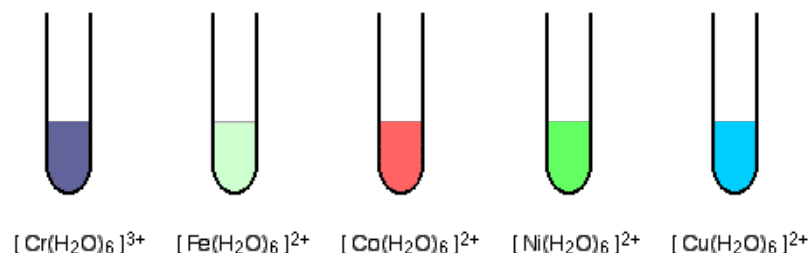
$\pi \rightarrow \pi^*$ (Allowed)
 $n \rightarrow \pi^*$ (Forbidden)

UV-visible spectra



3D Transitional metal ions

3	4	5	6	7	8	9	10	11	12
21	22	23	24	25	26	27	28	29	30
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Scandium 44.96	Titanium 47.88	Vanadium 50.94	Chromium 52.00	Manganese 54.94	Iron 55.85	Cobalt 58.93	Nickel 58.69	Copper 63.55	Zinc 65.39

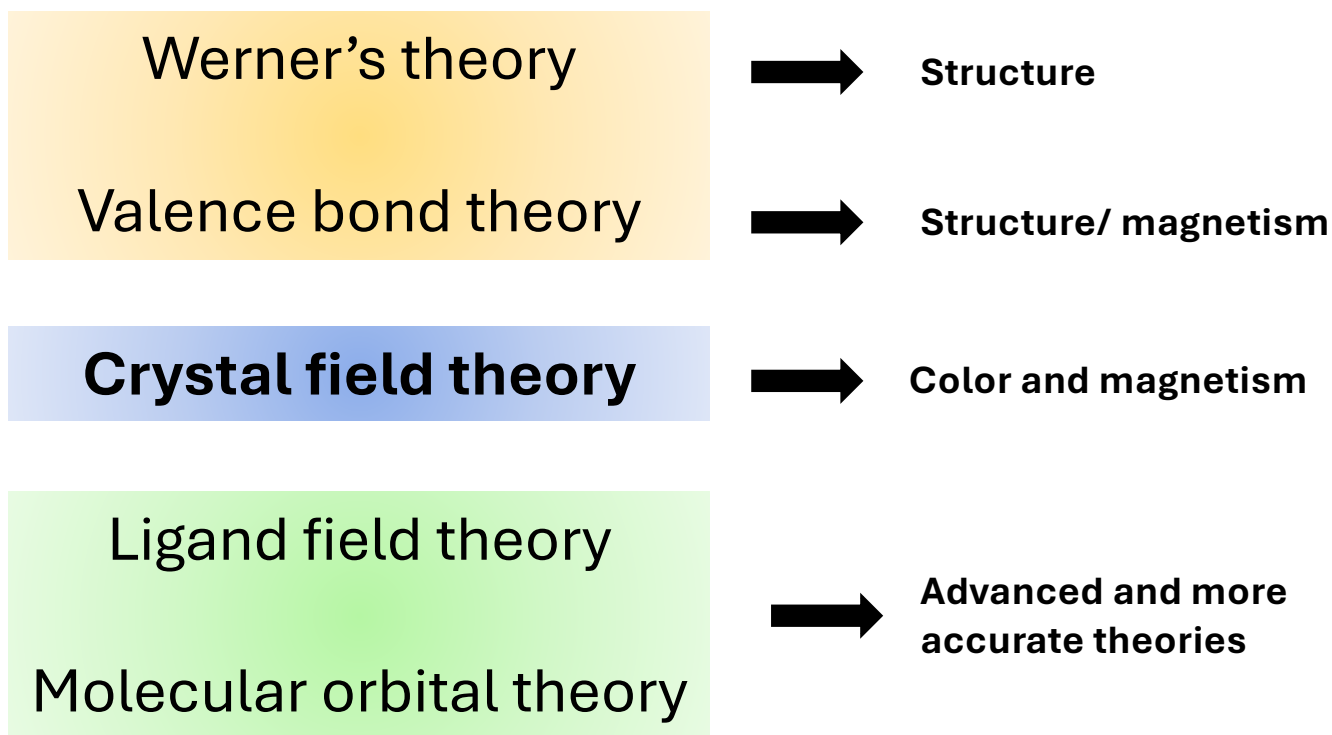


Why is aqueous solution of copper sulphate pale bluish in color while that of chromium ammine complex is violet?

Bonding in transitional metal complexes



Alfred Werner – first inorganic chemist to win the Nobel Prize (1913), developed the basis of modern coordination chemistry



Crystal field theory

Hans Bethe
J. H. van Vleck
(1930)

Concept: An electrostatic model treating ligands as point charges/dipoles interacting with metal d-orbitals, causing them to split in energy (crystal field splitting).

Key Ideas:

d-Orbital Splitting: Energy levels of d-orbitals (t_{2g} and e_g) split in octahedral/tetrahedral fields.

Color & Magnetism: Explains complex colors and high/low spin states based on ligand strength (spectrochemical series).

Assumptions of CFT

- 1) Pure electrostatic interactions – Think of metal complexes like an electric field
- 2) Ligands are point charges (no covalent interactions with the metal)

Negative ligand – ion, ion interaction

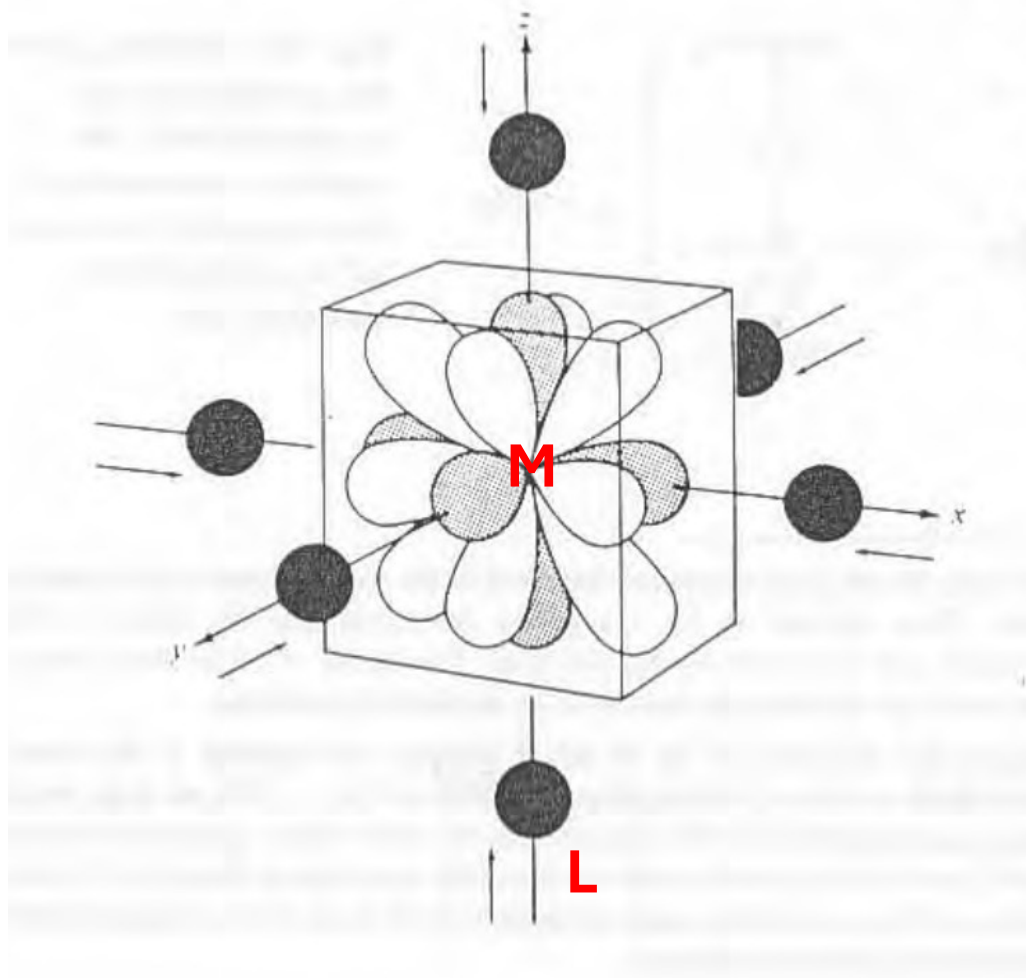
Neutral ligand – ion dipole interaction

- 3) Electrons from metals are repulsive from that of ligands
- 4) Electrons tend to occupy d-orbitals farthest from the ligands

Despite this rather unrealistic premise, the crystal field theory is quite successful in interpreting many important properties of complexes.

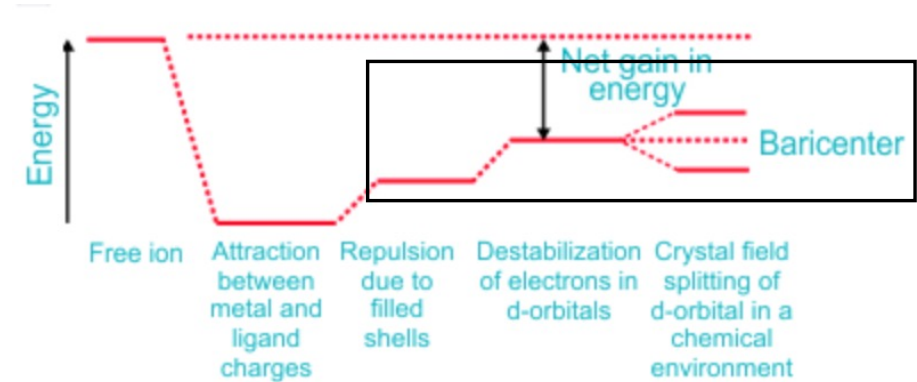
Huheey, Keiter Keiter

Ligands are approaching along the axes



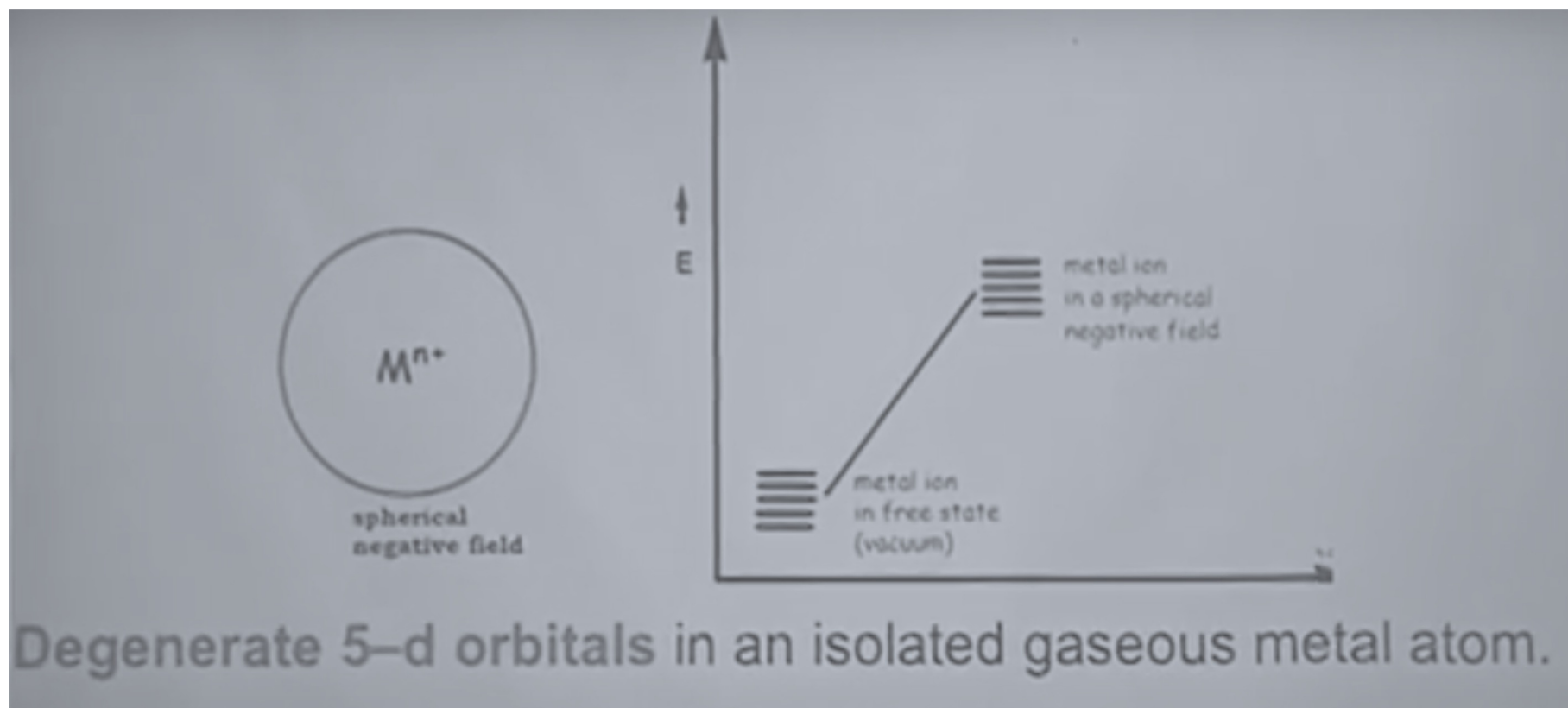
Octahedral (Oh) field

The Energy Profile

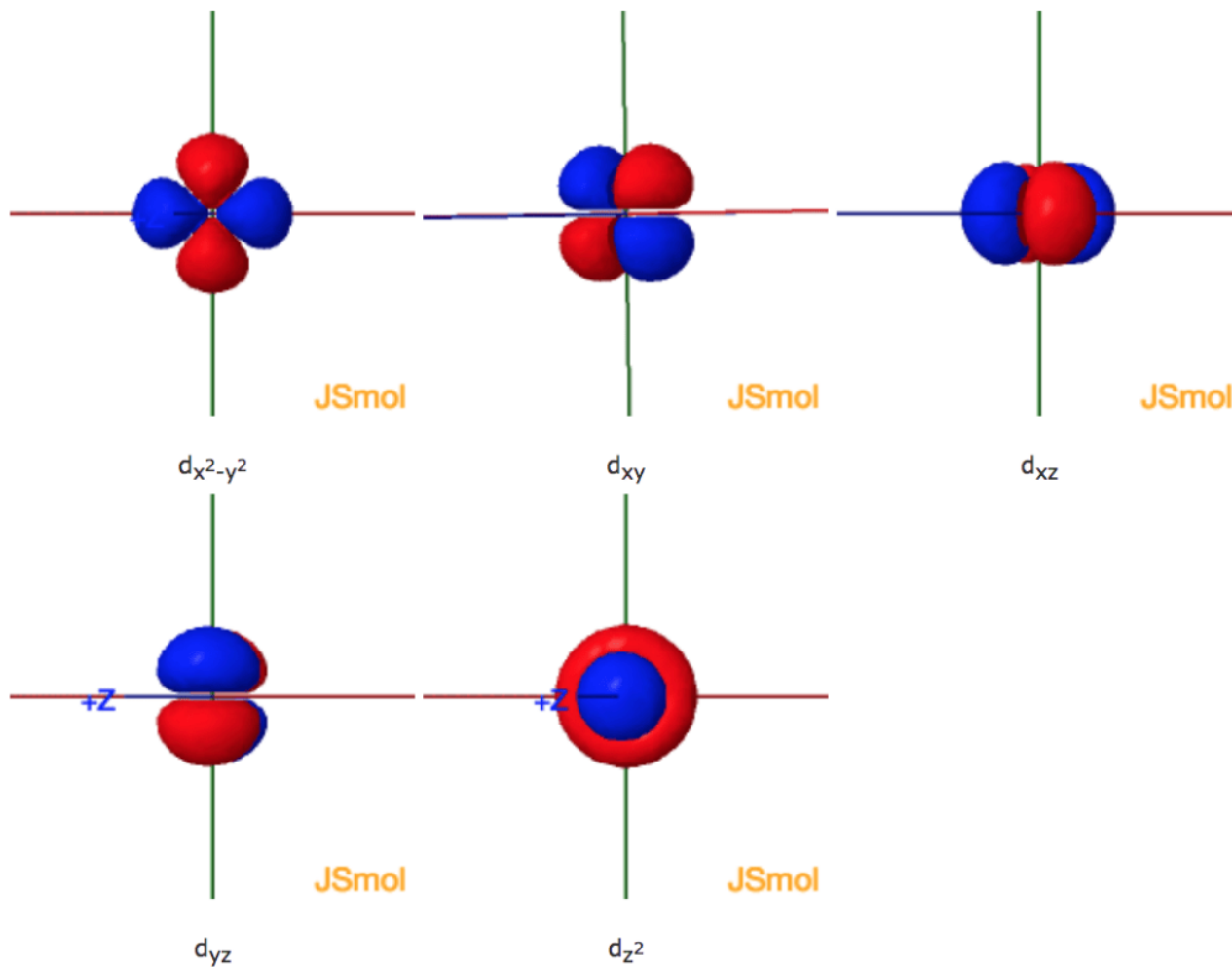


Electrons of metal is repulsed by electrons from the ligands

Considering a **Symmetrical ligand field**, all the d-orbitals will face similar kind of repulsion
Hence, all the d-orbitals energy will be raised, still they will remain degenerate

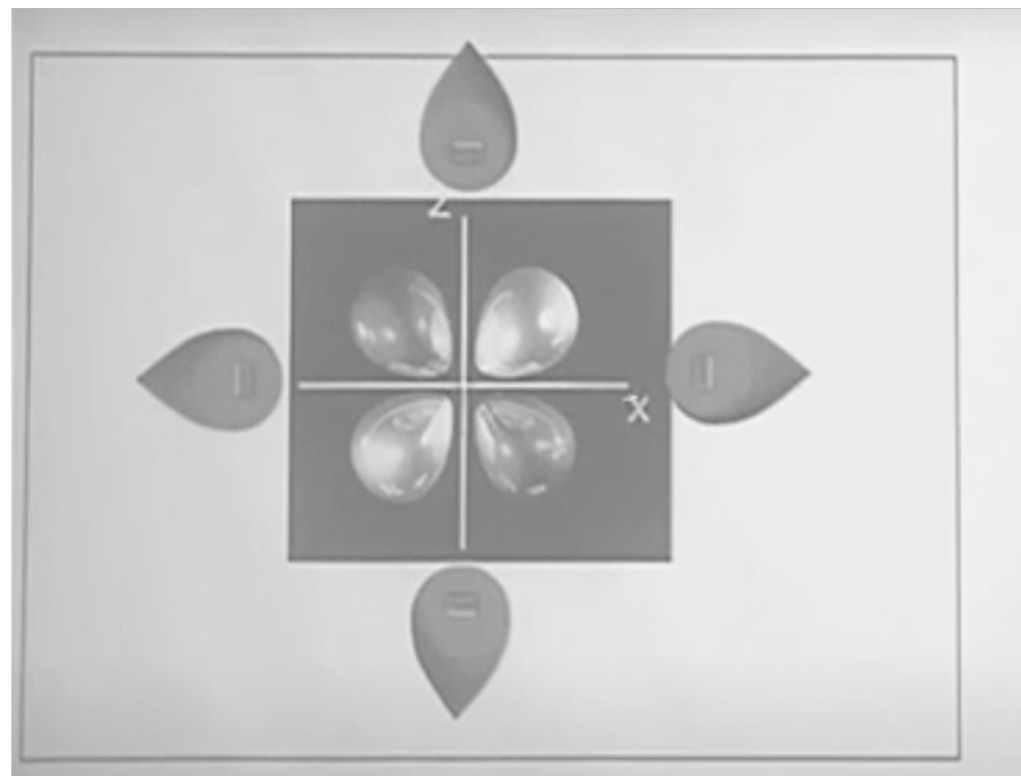


Shapes of d-orbitals are different

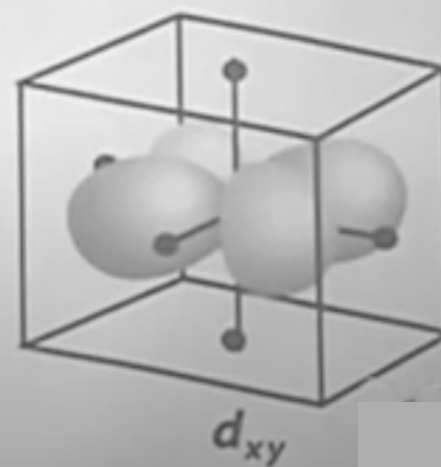
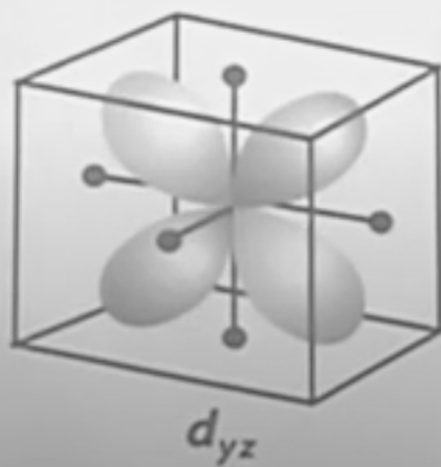
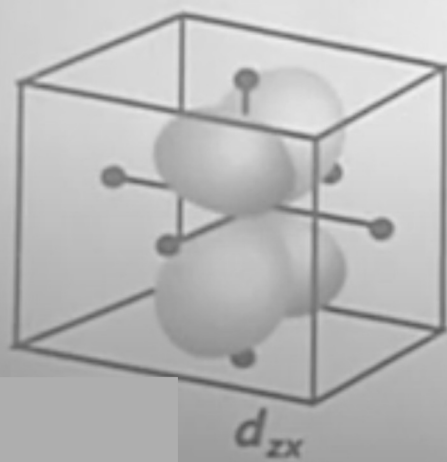
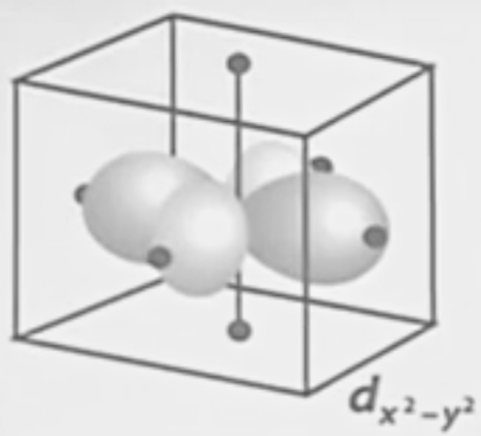
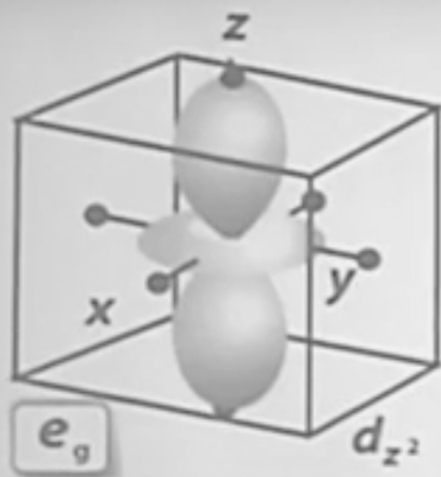


*3D animations
ChemTube 3D*

Ligands have face-to-face interaction with d_{z^2} and $d_{x^2-y^2}$ orbitals



Octahedral Complexes



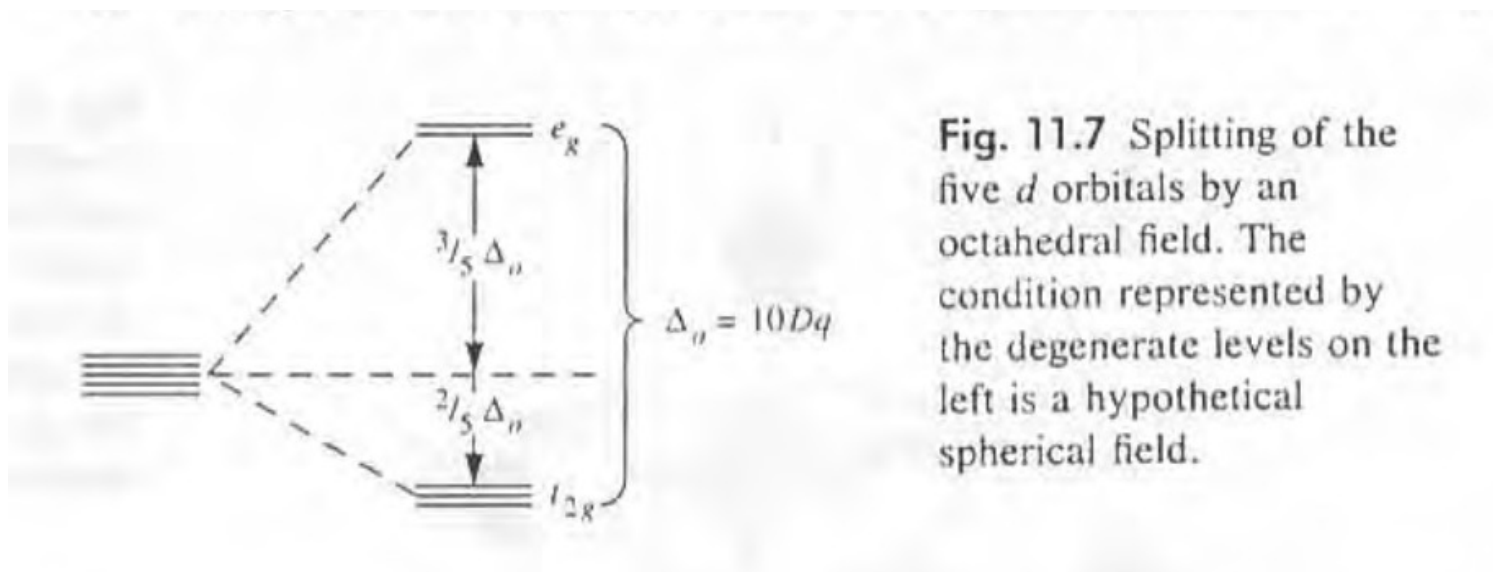


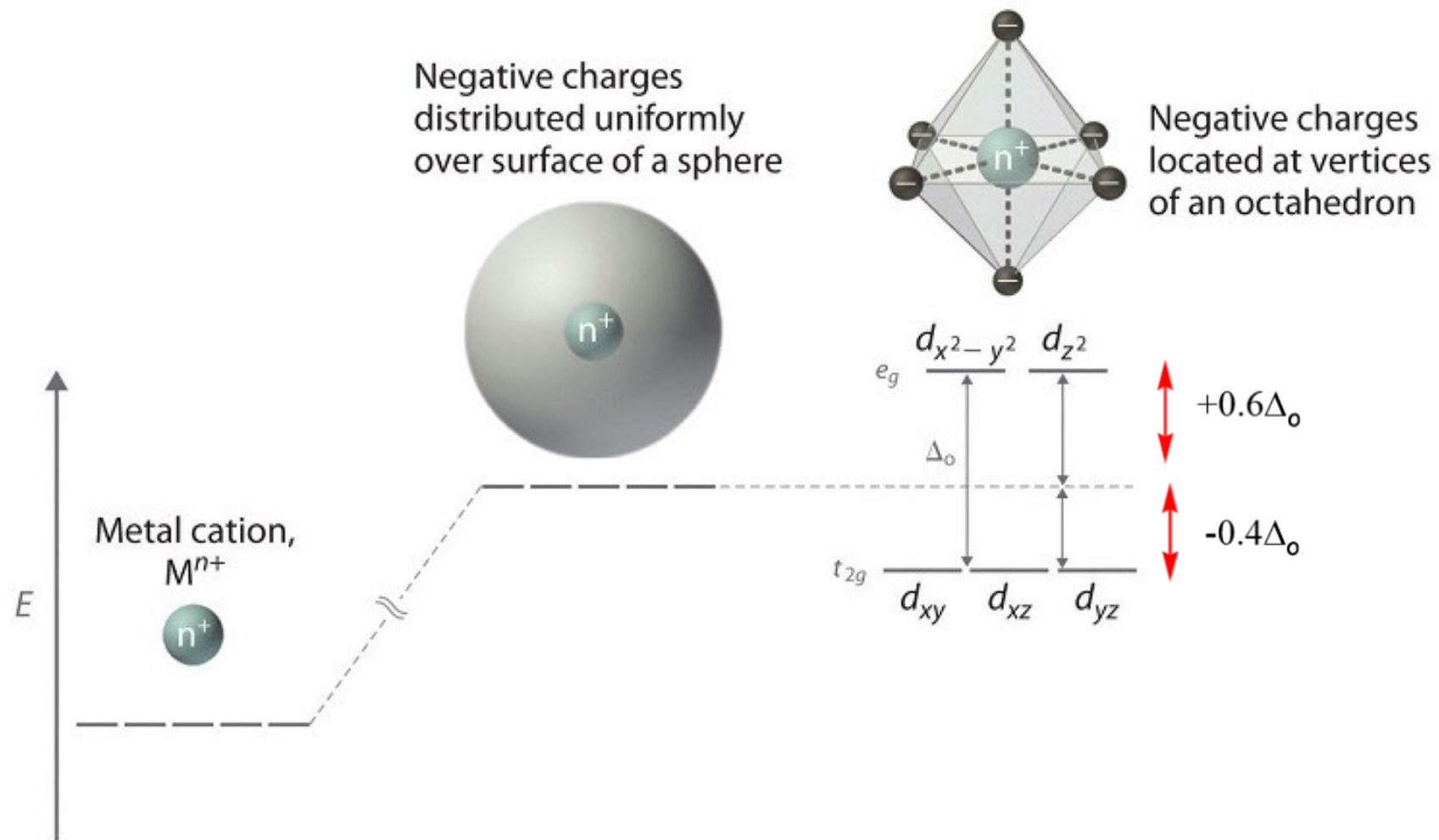
Fig. 11.7 Splitting of the five d orbitals by an octahedral field. The condition represented by the degenerate levels on the left is a hypothetical spherical field.

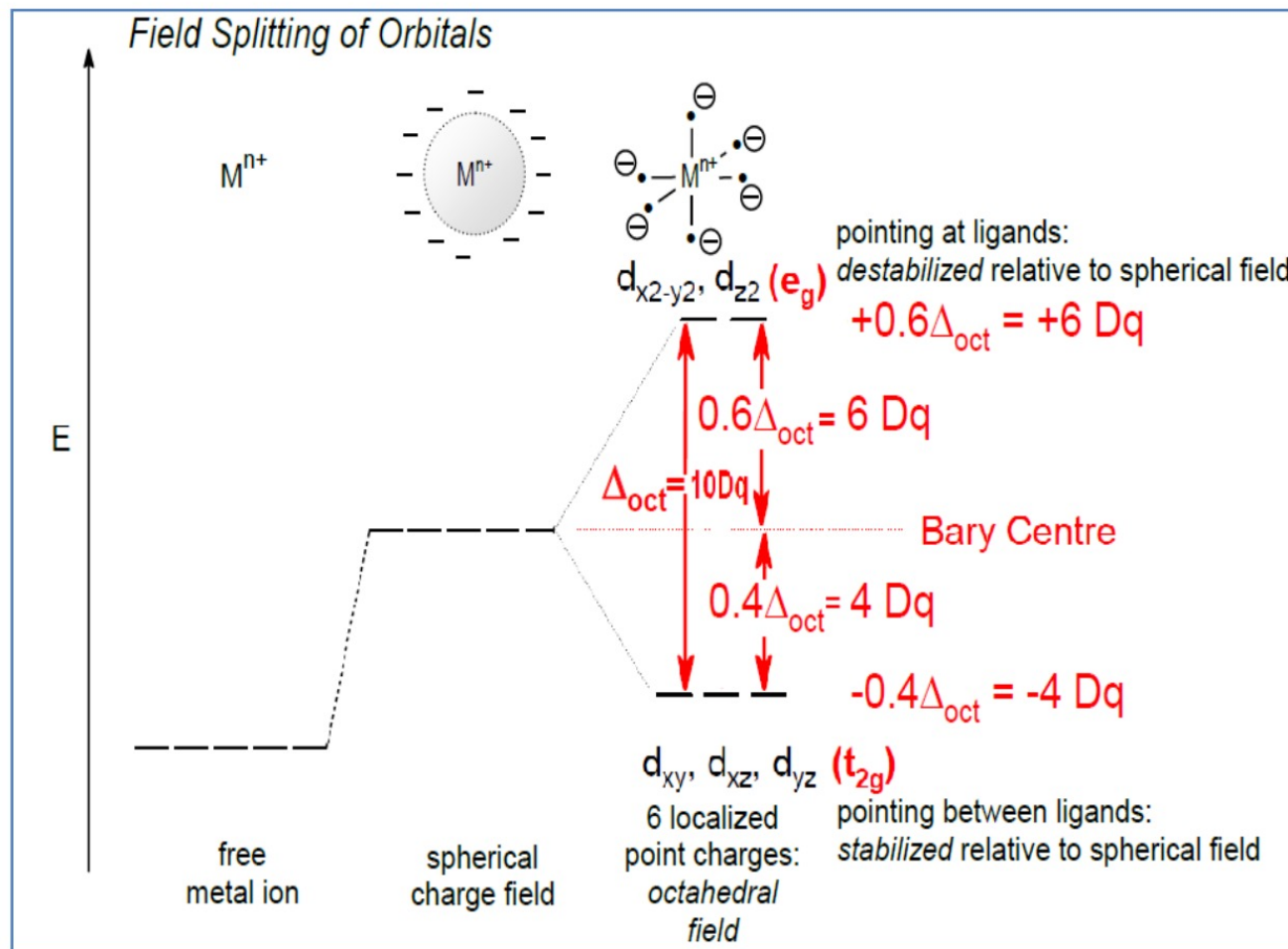
For O_h point group

x^2-y^2, z^2 orbitals: e_g

xy, xz, yz orbitals: t_{2g}

- Difference between t_{2g} and $e_g = \Delta_o$ or $10 Dq$.
- Conservation of barycenter from a spherical field to octahedral field indicates t_{2g} set must be **stabilized** as much as the e_g set is **destabilized**.



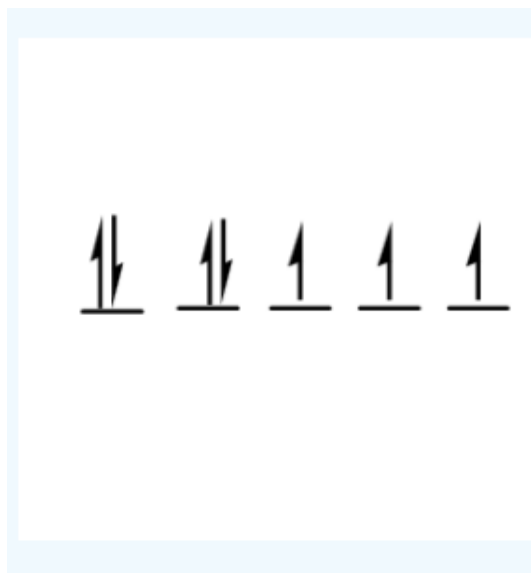


a, b = singly degenerate labels
 e = doubly degenerate
 t = triply degenerate

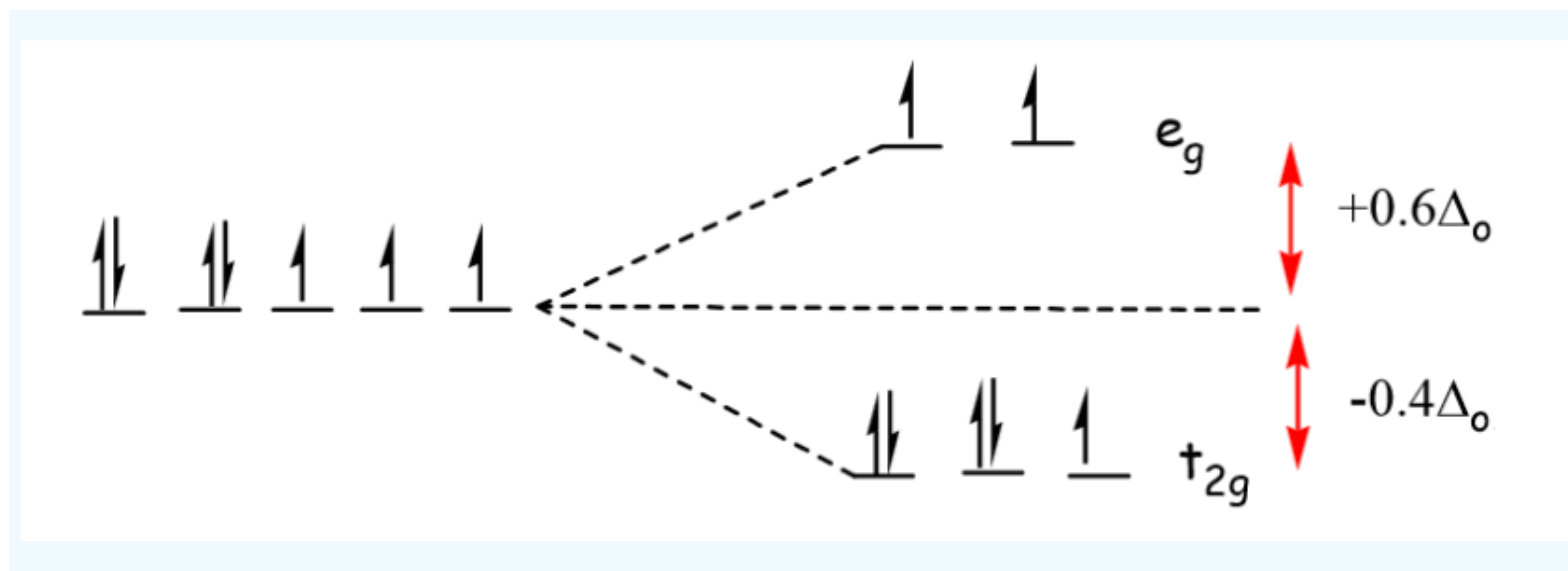
g = gerade (symmetrical about origin)
 u = ungerade (unsymmetrical about origin)

Figure2: Splitting of d-orbitals in Octahedral Field

Energy diagram of d7 in a symmetrical field where all d orbitals are degenerate



Energy diagram of d7 in a non-symmetrical field where all d orbitals lose degeneracy



Electronic Configuration

Atom	Configuration
Sc	[Ar] $4s^2 3d^1$
Ti	[Ar] $4s^2 3d^2$
V	[Ar] $4s^2 3d^3$
Cr	[Ar] $4s^1 3d^5$
Mn	[Ar] $4s^2 3d^5$
Fe	[Ar] $4s^2 3d^6$
Co	[Ar] $4s^2 3d^7$
Ni	[Ar] $4s^2 3d^8$
Cu	[Ar] $4s^1 3d^{10}$



First write the ground state configuration and then remove electrons

Mn²⁺

Mn: [Ar] $4s^2 3d^5$

Mn (0 state): [Ar] $4s^0 3d^7$

Mn (+2 state): [Ar] $4s^0 3d^5$